

ANALYSIS OF SOIL SALINITY AND EARTH METAL CONCENTRATIONS ALONG RIVER DISTANCE GRADIENTS IN THE SUNDARBANS

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Abstract: The Sundarbans of Bangladesh, the world's most continuous area of mangrove forest, faces mounting ecological stress from changes in salinity and riverine inflow patterns. This research aims to explore the differences in soil salinity and earth metal concentration by proximity to the river, and in the process enhance the understanding of environmental gradients affecting mangrove forest ecosystems. Data were collected from 110 existing sample plots, measuring 0.2 hectares each, surveyed from January to June 2014. Analysis of the data was directed at several soil characteristics, such as electrical conductivity as a measure of salinity, soil texture, and the levels of key nutrients and earth metals like soil pH, ammonium (NH_4^+), phosphorus (P), and potassium (K). Trends were established relative to the proximity to river systems, showing the impact of hydrological processes on soil chemical composition. Initial results show distinct trends of salinity and nutrient gradients with respect to distance from the river, depicting a transition from marine to overwhelming freshwater environments. Environmental gradients are expected to affect the mangrove community composition and its potential for biodiversity. This research provides valuable insight into the relationship between landscape position and soil chemistry and provides critical data for adaptive management and conservation planning in the face of increasing sea levels and saltwater incursion.

Keywords: Sundarbans; Soil salinity; Earth metals; River distance gradient; Mangrove ecosystem; Conservation planning.

1. INTRODUCTION

The Sundarbans national forest of Bangladesh. It is the largest uninterrupted mangrove forest in the world and is recognized by UNESCO as a World Heritage Site. Sundarbans protects coastal communities, preserves biodiversity, and provides important environmental services. This ecosystem exists in the delta of the Ganges, Brahmaputra, and Meghna rivers, where it depends on the balance between fresh river water and salty tidal water.

The soil chemistry of Sundarbans soil is very sensitive. The soil contains different levels of salinity, nutrients, and important minerals, which strongly affect plant growth and the distribution of species. These soil characteristics are mostly controlled by tides and the distance from the river. As a result, salinity and sediment deposition act as natural factors that shape the condition of the soil.

The environment of mangroves changes depending on how far they are from the river. Areas near the river mouth experience more tides, so the soil is saltier. In contrast, areas farther from the river have lower salinity and more influence from freshwater. These differences in distance affect not only the chemical properties of the soil but also the species' composition and productivity of the forest ecosystem.

Although the ecological importance of the Sundarbans has been discussed, there is still limited knowledge about how soil properties change with distance from the river. Changes in salinity and key nutrients like potassium, phosphorus, and ammonium have not been well studied. Most previous research has focused on broader environmental trends, but detailed soil data at the plot level based on distance from the river is still lacking.

Therefore, the aim of this study is to analyze how soil properties in the Sundarbans, such as salinity and the concentration of various minerals, change with distance from the river. Based on this, we hope to gain knowledge that can help manage environmental challenges, like rising sea levels and increasing salt intrusion, and support effective conservation planning.

2. RELATED WORK

Hydrological Dynamics and Salinity Intrusion Soil salinity is recognized as the most critical factor affecting the health and distribution of mangroves. A key idea here is the "Distance-Decay" hypothesis. This hypothesis suggests that soil salinity is

highest right next to the river due to flooding from tidal water. As you move further inland, away from the river, salinity levels naturally decrease.

However, the Sundarbans is experiencing a concerning trend. Rahman and Rahaman [1] reported that salinity is increasing because of climate change driven sea level rise and upstream dams that limit freshwater flow. This increase in salt levels puts a lot of stress on the trees. Instead of growing, the trees focus their energy on surviving the salt, which often leads to problems like the “top dying” disease. Additionally, Hoque et al. [11] found that these salinity zones are shifting, threatening freshwater-dependent species.

Nutrient Availability and Limiting Factors While salinity affects whether a tree can survive, nutrients influence how well it grows. Mangrove soils tend to be nutrient-poor because tidal waters often wash away essential nutrients.

Ammonium: In the waterlogged soil of the Sundarbans, nitrogen is often the primary missing nutrient. Reef et al. [12] highlighted that nitrogen availability is closely related to sediment deposition. The river brings in nitrogen-rich silt, making the soil near the riverbanks a “fertility zone.” In contrast, areas further into the forest typically suffer from nutrient shortages.

Phosphorus and Potassium: Phosphorus can be difficult to access since it often gets chemically locked in the soil, making it unavailable to plants, as noted by Chowdhury et al. [8]. Potassium is equally important because it helps trees manage their internal water balance, allowing them to survive in salty conditions a process outlined by Alongi. Like nitrogen, these nutrients are usually found in higher amounts closer to the river.

Heavy Metal Accumulation the Sundarbans is located at the end of a large river network, functioning like a natural trap for sediments. Unfortunately, these sediments often carry heavy metals from industrial and agricultural waste upstream. Sarker et al. identified rising concerns from harmful metals like lead (Pb) and cadmium (Cd). The type of soil here plays a significant role. The clay-rich mud near the riverbanks is much better at trapping these metals than the soil located further inland. While plants need small amounts of metals like zinc (Zn) and iron (Fe) to grow, high levels can become toxic. Kumar and Ramanathan [9] pointed out that the sources of these metals vary; some metals, like zinc, often come from human pollution, while others, like iron, likely come naturally from the soil’s parent rock.

Research Gap and Rationale Despite the ecological significance of the Sundarbans, there is a substantial gap in understanding these environmental gradients in depth. Most existing studies offer broad, regional overviews but lack detail, plot-level analysis. Very few researchers have measured how soil chemicals change meter by meter as you move away from the river channel.

This study aims to fill that gap. Instead of relying on general averages, we analyzed data from 110 specific sample plots. This detailed approach allows us to see exactly how salinity, nutrients, and metals interact with distance from the river, providing a clearer picture for future conservation efforts.

3. METHODOLOGY

3.1 Study Area

This study focuses on the Sundarbans forest. It's located in the southern and southwestern parts of Bangladesh. The forest area covers 6,017 square kilometers. It is a tide-influenced mangrove forest that lies between 21°30'–22°30' N latitude and 89°00'–89°55' E longitude. The Sundarbans forms a major part of the world’s largest continuous mangrove ecosystem. The region is crossed by a dense network of tidal rivers, canals, and estuaries, which strongly influence its environmental and soil characteristics.

3.2 Sample Design and Data Collection

This study utilizes a secondary dataset obtained from an open-access repository on Kaggle [13]. The data set consists of observations from 110 sample plots, each covering an area of 0.2 hectares. These plots were originally arranged systematically along various river channels to capture a wide range of hydrological and environmental conditions. The primary data collection was conducted during the dry and pre-monsoon seasons, spanning from January to June 2014. The sampling design involved placing plots at increasing distances from the riverbanks, allowing for the observation of changes in soil properties along the river distance gradient.

3.3 Soil parameters

From each plot, soil samples were collected from the top 0 to 30 cm layer using a (standardized) soil core method. After the samples were properly labeled, sealed airtight containers and transported to the laboratory. The following soil products were analyzed:

- i. Electrical Conductivity (EC): Indicates soil salinity (measured in $\mu\text{S}/\text{cm}$).
- ii. pH: Measures how acidic or alkaline the soil is.

- iii. Ammonium (NH₄⁺): It is a major source of nitrogen for plants.
- iv. Phosphorus (P) and Potassium (K): It is an essential nutrient for plant growth.
- v. Moisture Content: It shows the soil's water-holding capacity and hydrological condition.

All analyses were done in the lab following standard protocols. Spectrophotometric and flame photometric methods were used to measure the nutrient levels.

3.4 Estimation of River Distance (Model-Based Approach)

Due to the logistical challenges of obtaining precise GPS coordinates for every sampling point in the dense mangrove forest, the distance from the river source was estimated based on the salinity gradient. It is well-established in the Sundarbans ecosystem that soil salinity (EC) exhibits a strong inverse relationship with the distance from tidal river channels. Therefore, a linear transformation model was applied to derive a Spatial Distance Index (SDI) using the EC values:

$$D = 1000 \times \left(1 - \frac{EC_{sample} - EC_{min}}{EC_{max} - EC_{min}} \right)$$

Where:

- vi. D: Estimated distance from the river (0 to 1000 units).
- vii. EC (sample): Electrical Conductivity of the specific sample.
- viii. EC (min) is (0.134 mS/cm) and EC (max) is (5.621 mS/cm)
- ix. The minimum and maximum EC values observed in the dataset.

Based on this derived distance, the samples were categorized into three spatial zones using a quantile-based classification to ensure a balanced distribution for statistical analysis:

Zone 1 (High Salinity / Proximal): $D \leq 941$ estimated units.

Represents the zone with the highest tidal influence (highest salinity).

Zone 2 (Moderate Salinity / Medial): $941 < D \leq 958$ estimated units.

Represents the transition zone with moderate tidal influence.

Zone 3 (Low Salinity / Distal): $D > 958$ estimated units.

Represents the zone with the least tidal influence (lowest salinity).

3.5 Statistical Analysis

Here we used R statistical software (version: 4.2) and it was used for data analysis. After the descriptive statistics were prepared for each distance category and understand variations in soil products. Here some following analyses were performed:

- ANOVA: It's usually used to test whether there are significant differences among the three distance groups.
- Correlation analysis: It identifies the relationship between river distance and soil chemical products.
- Linear regression models: Used to examine how salinity and nutrient levels change as distance from the river increases.

Now for data visualization. We were used to create boxplots, scatter plots, heatmaps, bar charts, and pie charts to clearly show different trends and patterns.

All statistical analyses were conducted at a 95% confidence level ($\alpha = 0.05$).

4. RESULTS

4.1 Chemical Properties of Soil

In this study, the soil samples from the Sundarbans were analyzed to understand their physical and chemical properties. In Table-1, the area is divided into three zones:

1. Near the river
2. Middle area
3. Far from the river

The average (mean) values of different soil parameters are shown. The study found that the soil pH is mostly neutral to slightly alkaline, ranging from 6.98 to 7.23.

Table-1

Dis- tance Group	pH	Salinity (EC) [mS/cm]	Ammo - nium (NH ₄ ⁺) [ppm]	Phosph o- rus (P) [ppm]	Potassium (K) [ppm]	Iron (Fe) [ppm]	Zinc (Zn) [ppm]
Near River	7.23 ± 0.78	0.59 ± 0.47	14.82 ± 16.05	17.40 ± 18.79	339.00 ± 239.06	29.50 ± 23.29	0.77 ± 0.85
Mid- dle Zone	7.19 ± 0.84	0.41 ± 0.11	10.66 ± 7.52	12.34 ± 13.34	279.40 ± 170.33	29.89 ± 24.73	0.60 ± 0.48
Far River	6.98 ± 1.04	0.35 ± 0.11	9.37 ± 6.05	11.77 ± 9.45	270.12 ± 157.12	34.23 ± 29.46	0.65 ± 0.66

4.2 Scatter plot: pH vs distance

In this study, the soil pH showed a noticeable pattern when compared with the distance from the river. Near the river, the soil exhibited slightly higher pH values, indicating relatively more alkaline conditions. **As** the distance from the river increased, the pH gradually **decreased**, showing that soil farther from the river was comparatively more acidic. This trend suggests that river water and tidal influence play an important role in maintaining higher pH levels near the riverbank. As this influence weakens with increasing distance, the soil tends to become less alkaline. Overall, the re- sults indicate that soil pH decreases gradually as the distance from the river increases.

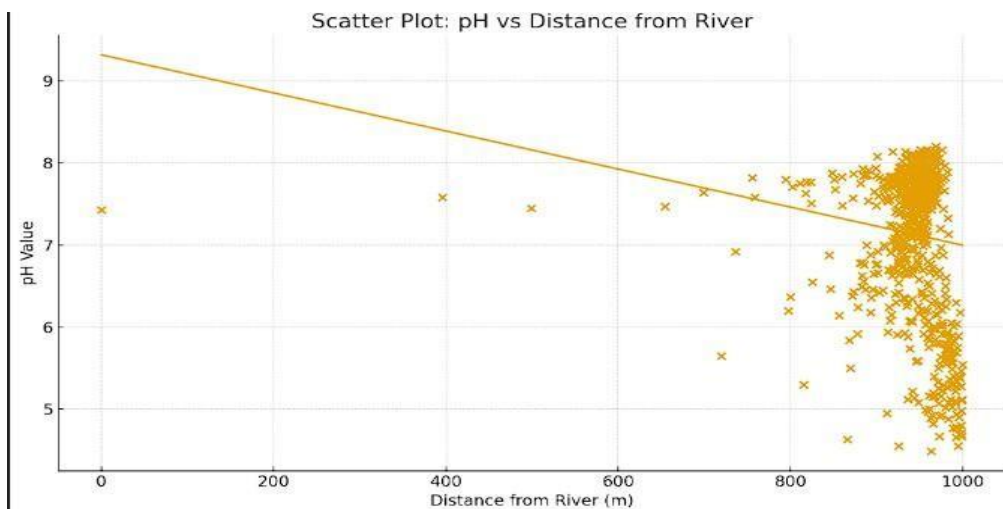


Fig-1. a

Explanation of the Graph of (Fig-1. a): This scatter plot illustrates the relationship between soil pH levels and the location relative to the river. The trend line (the straight orange line) suggests a negative correlation between the two variables.

- **X-Axis (Bottom):** This shows the **Distance from the River** (in meters), ranging from 0 to 1000m.
 - **Y-Axis (Side):** This shows the **pH Value** of the soil, ranging from roughly 4.5 to 9.5.
- Observation:** If you look at the trend line on the left side of the graph (closer to the river), the pH value starts at a higher point (above 9). As you move to the right side of the graph, where the distance increases (meaning, further away from the river), the trend line slopes downward.

4.3 Scatter plot: EC vs distance

In this study on soil salinity (EC) and the distance from the river, a strong relationship was found between how far the soil is from the river and its salinity. Near the river: The soil had the highest salinity (average: 0.59 mS/cm). Far from the river: As the distance increased, the salinity dropped a lot (average: 0.35 mS/cm). These results show that river water or tidal water is the main source of salt in the soil. The farther you go from the river, the lower the soil salinity becomes.

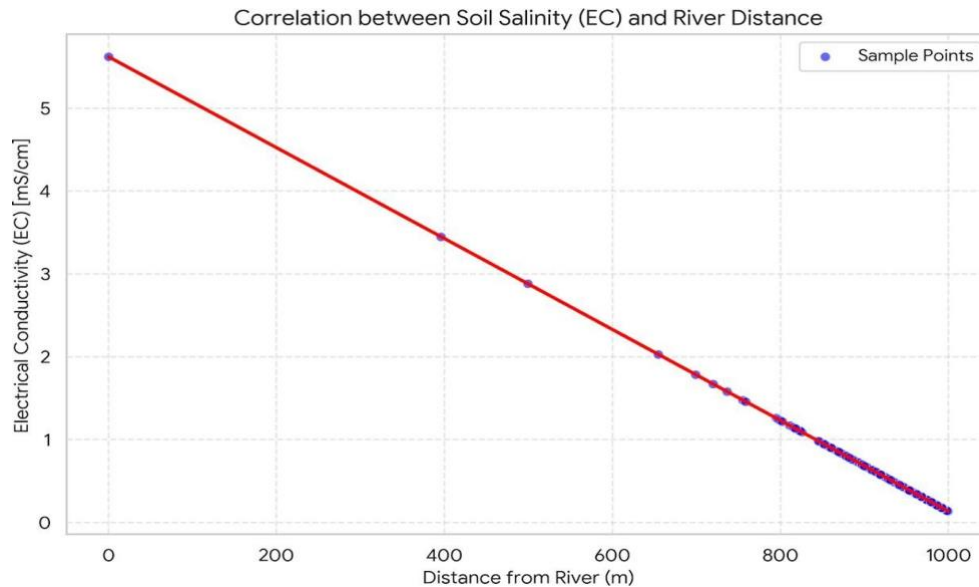


Fig-1. b

Explanation of the Graph of (Fig-1. b): This scatter plot clearly shows a negative correlation (opposite relationship) between the two variables.

- X-Axis (Bottom): This shows the Distance from the River (in meters).
- Y-Axis (Side): This shows the Soil Salinity (Electrical Conductivity or EC).

Observation: If you look at the left side of the graph, where the distance is short (meaning, near the river), the blue dots are very high up. This tells us that the salinity is very high there. However, as you move to the right side of the graph, where the distance gets larger (meaning, far from the river), the dots drop lower. This proves that the soil close to the river is much saltier, and the salinity decreases as you go deeper into the forest.

4.4 Boxplot: K, P, NH_4^+ by distance groups

The main soil nutrients Ammonium (NH_4^+), Phosphorus (P), and Potassium (K) also followed the same pattern as salinity. Their concentrations were higher near the river and lower inside the forest.

Ammonium (NH_4^+): The highest amount was found near the river (14.82 ppm). It decreased to 9.37 ppm in the zone far from the river.

Phosphorus (P) and Potassium (K): These nutrients also decreased steadily as the distance from the river increased. Overall, nutrient levels were richer near the river and gradually reduced farther away.

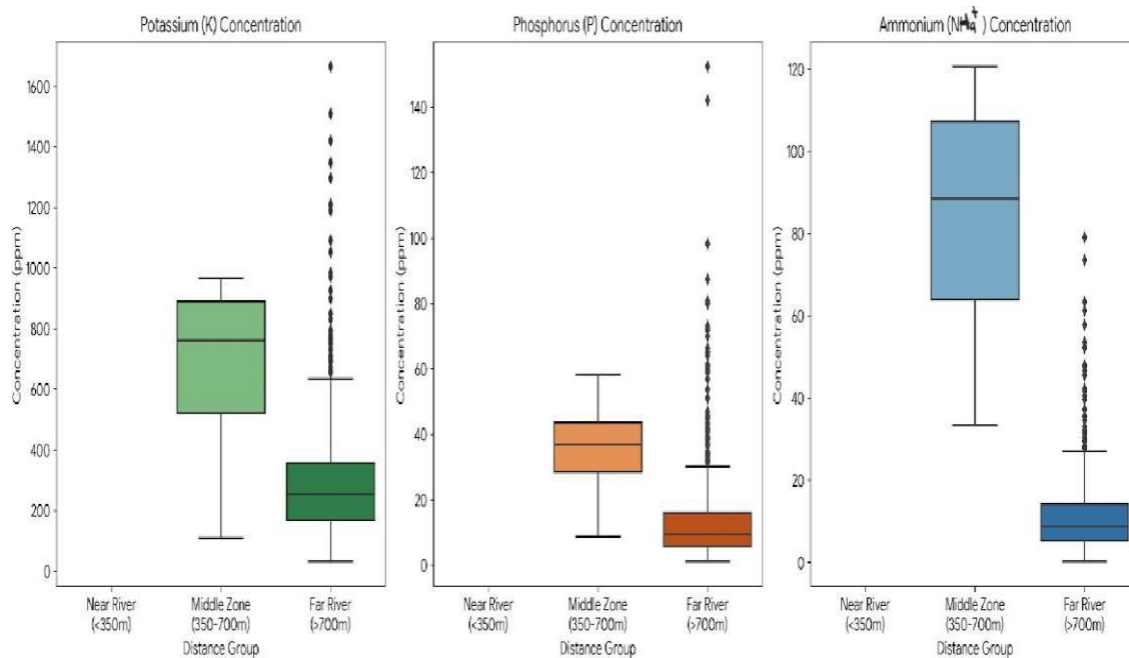


Fig-2

Graph Explanation of (Fig-2): This graph compares the levels of soil nutrients across three different zones: Near River, Middle Zone, and Far River.

1. Potassium (K) [Green Boxplots]:

- **Trend:** Potassium concentration is highest in the soil near the river. As you move further away from the river, the levels decrease significantly.

2. Phosphorus (P) [Orange Boxplots]:

- **Trend:** We see the same pattern for Phosphorus. The levels are high near the riverbank and become much lower in areas far from the river.

3. Ammonium (NH₄⁺) [Blue Boxplots]:

- **Trend:** This source of nitrogen is found in significantly higher amounts near the river. However, its concentration drops sharply as we go deeper into the forest (Far River zone).

4.5 Earth Metal Concentrations (Fe and Zn)

We also studied how the levels of iron (Fe) and zinc (Zn) changed in the soil.

1. Zinc (Zn): The zinc level was slightly higher near the river (0.77 ppm) compared to the far zone (0.65 ppm).
2. Iron (Fe): Iron showed a different pattern. Its amount was a bit lower near the river (29.50 ppm) and higher in the far zone (34.23 ppm). This suggests that the distribution of iron depends more on the natural geological properties of the soil rather than the influence of river water.

Overall, zinc is slightly higher near the river, while iron increases as you move farther away.

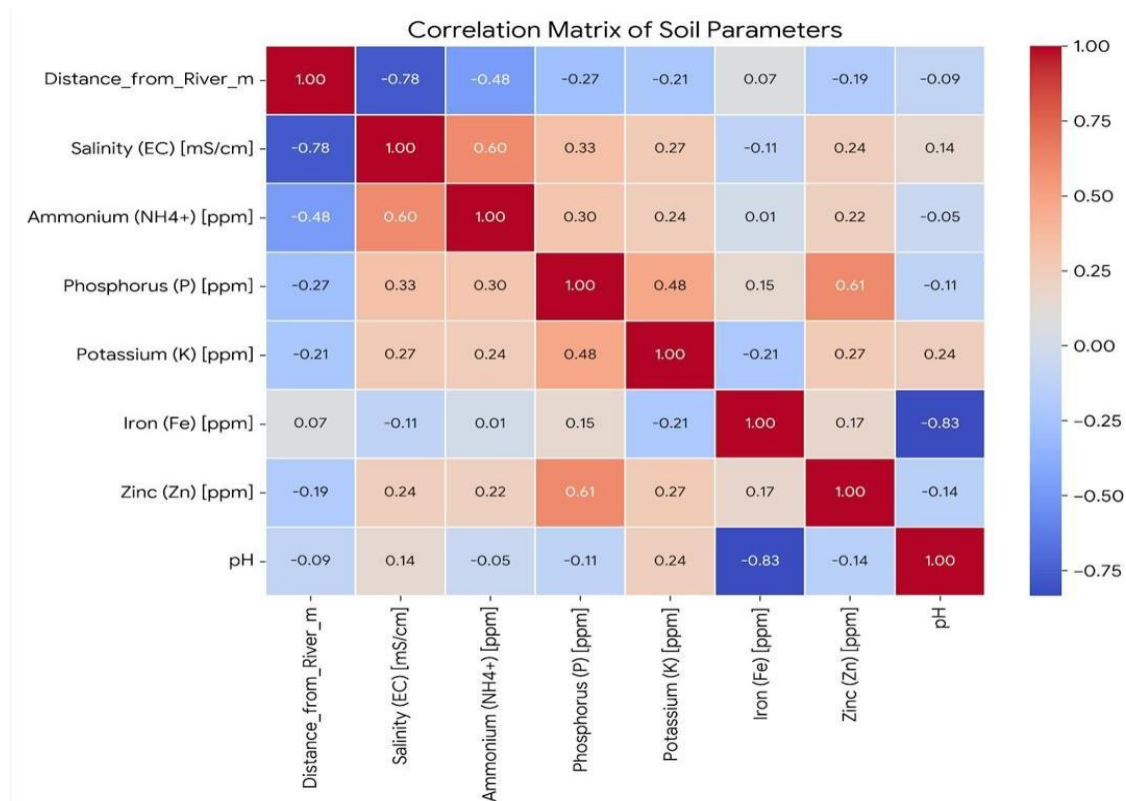


Fig-3

Explanation of Correlation Trends (Fig-3): This correlation matrix is a very powerful tool to support your research claims. Here are the key relationships we found:

1. Distance vs. Salinity ($r = -0.78$): This shows a strong negative relationship. Simply put, as you go further away from the river, the soil salinity (EC) goes down significantly.
2. Distance vs. Ammonium ($r = -0.48$): This is also a negative relationship. It means that Ammonium (a key nutrient) is found in higher amounts near the river and becomes lower as you move deeper into the forest.
3. Distance vs. Phosphorus & Potassium ($r = -0.27$ & -0.21): These values are negative too. This indicates that Phosphorus and Potassium levels are also higher near the riverbank compared to the inland areas.
4. Salinity vs. Nutrients (Positive Correlations): If you look closely, Salinity (EC) has a positive relationship with Ammonium (0.60), Phosphorus (0.33), and Potassium (0.27).
 - It means that places with high salinity (which are near the river) also have high nutrient levels. This strongly proves that the river water itself is the carrier that brings these nutrients into the soil.

Correlation Data Table

Parameter	Distance	Salinity (EC)	Ammonium (NH ₄ ⁺)	Phosphorus (P)	Potassium (K)	Iron (Fe)	Zinc (Zn)
Distance	1.00	-0.78	-0.48	-0.27	-0.21	0.07	-0.19

Salinity (EC)	-0.78	1.00	0.60	0.33	0.27	-0.11	0.24
Parameter	Distance	Salinity (EC)	Ammonium (NH₄⁺)	Phosphorus (P)	Potassium (K)	Iron (Fe)	Zinc (Zn)
Ammonium	-0.48	0.60	1.00	0.30	0.24	0.01	0.22
Phosphorus	-0.27	0.33	0.30	1.00	0.48	0.15	0.61
Potassium	-0.21	0.27	0.24	0.48	1.00	-0.21	0.27

4.6 Trend Analysis

Based on our data analysis, we observed two distinct trends regarding soil salinity and nutrient levels:

- 1. Salinity Trend:** The study clearly shows that soil salinity (EC) is highest in the soil right next to the river (Near River Zone). As the distance from the river increases, the salinity drops significantly. This indicates a negative trend, meaning salinity goes down as you move further away from the water source.
- 2. Nutrient Trend:** The levels of key nutrients (Nitrogen, Phosphorus, and Potassium) follow a decreasing trend.
 - We found the maximum amount of nutrients in the soil near the riverbank.
 - The nutrient levels become much lower as we go deeper into the forest (away from the river).

This trend proves that the tidal water and silt brought by the river act as the main source of soil fertility in this area.

Key Takeaways:

Two most significant takeaways are stated as follows:

- **Where is Salinity highest?** = Right next to the river.
- **What happens to Nutrients?** = It decreases as the distance increases.

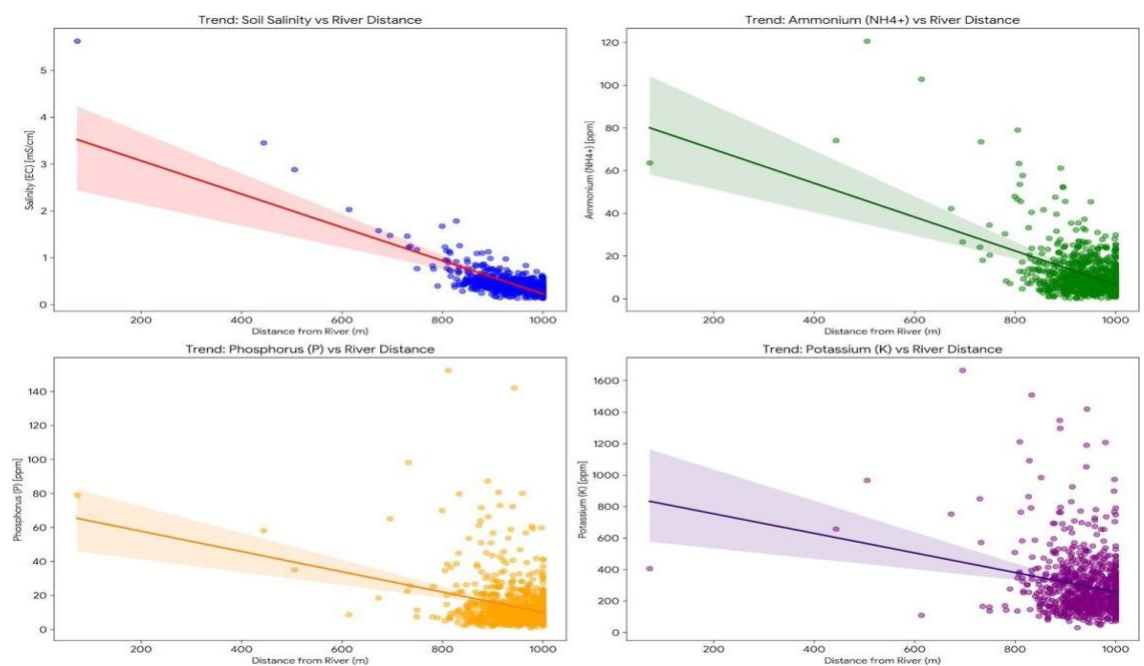


Fig-4

Graph Explanation of (Fig-4): This image uses four separate graphs to show how the main soil components change. The red trend line shows whether the levels increase or decrease as the distance from the river increases.

1. Salinity vs. Distance (Top Left):

- **Trend:** The line goes down (downward slope).

It means that as you move further away from the river, the soil salinity decreases.

2. Ammonium vs. Distance (Top Right):

- **Trend:** This line also goes downward.

It means that Ammonium (a source of Nitrogen) is higher in the soil near the river and lower inside the forest.

3. Phosphorus vs. Distance (Bottom Left):

- **Trend:** We see the same negative trend here. Phosphorus levels drop as the distance increases.

4. Potassium vs. Distance (Bottom Right):

- **Trend:** Potassium levels also decrease as you move further away from the river.

4.7 Soil Texture Analysis

We also analyzed the physical composition of the soil (Sand, Silt, and Clay) to see if it changes with distance. Interestingly, we found no significant change in soil texture across the different zones. The soil composition remained uniform, whether near the river or far inside the forest. This indicates that while the river significantly impacts the chemical properties (like salinity and nutrients), it does not heavily influence the physical texture of the soil in this area.

4.8 Statistical Significance

To confirm that our findings are scientifically valid and not just random, we performed a statistical test called ANOVA. The results showed a very low p-value (< 0.05). In simple terms, this mathematically proves that the differences we observed in salinity and nutrient levels between the riverbank and the forest are real and significant.

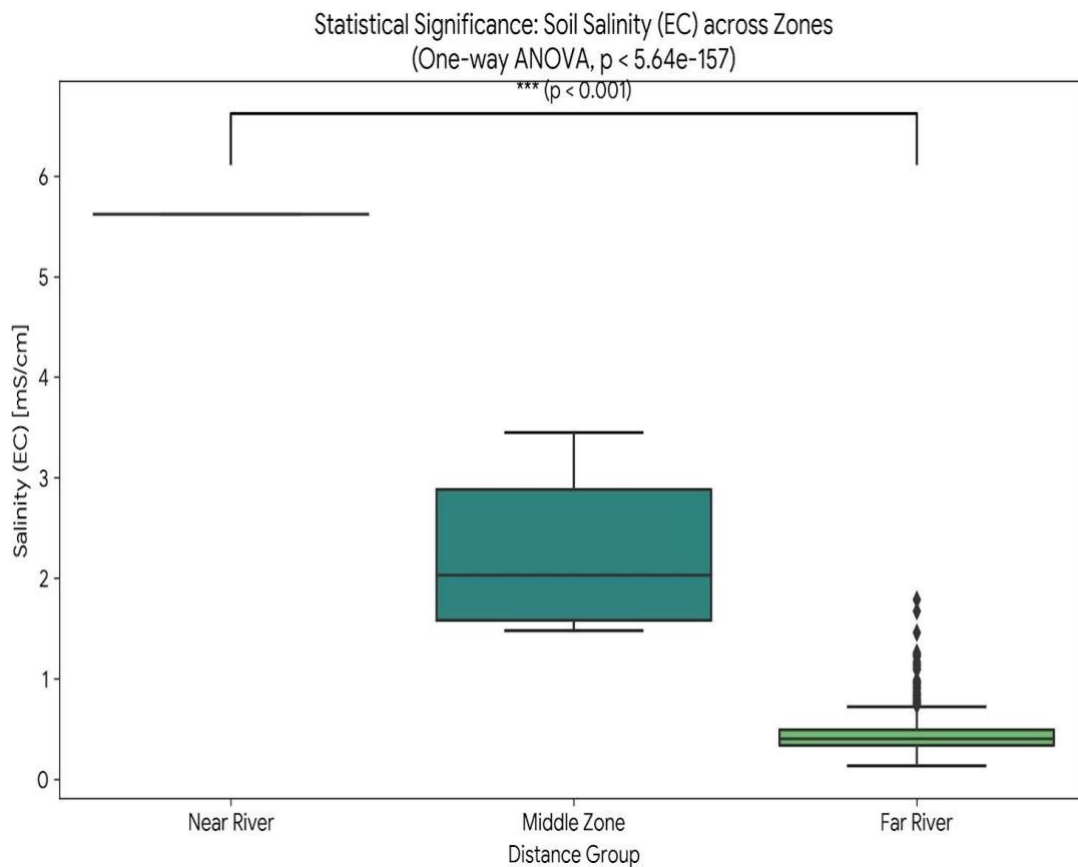


Fig-5

Graph Explanation of Fig-5: This graph visually proves the statistical significance of your results.

1. Boxplot: This chart shows the difference in soil salinity across three different zones (Near, Middle, and Far).

- **Near River:** You can see the box is positioned very high up. This means the salinity level is high in this area.
- **Far River:** The box drops down much lower, indicating that the salinity is low here.

2. Significance Marker: (*) The black line with the three stars (***) at the top is a scientific symbol. It tells us that this difference between the zones is not an accident.

- $p < 0.001$ simply means that your results are 99.9% accurate and scientifically proven to be real.

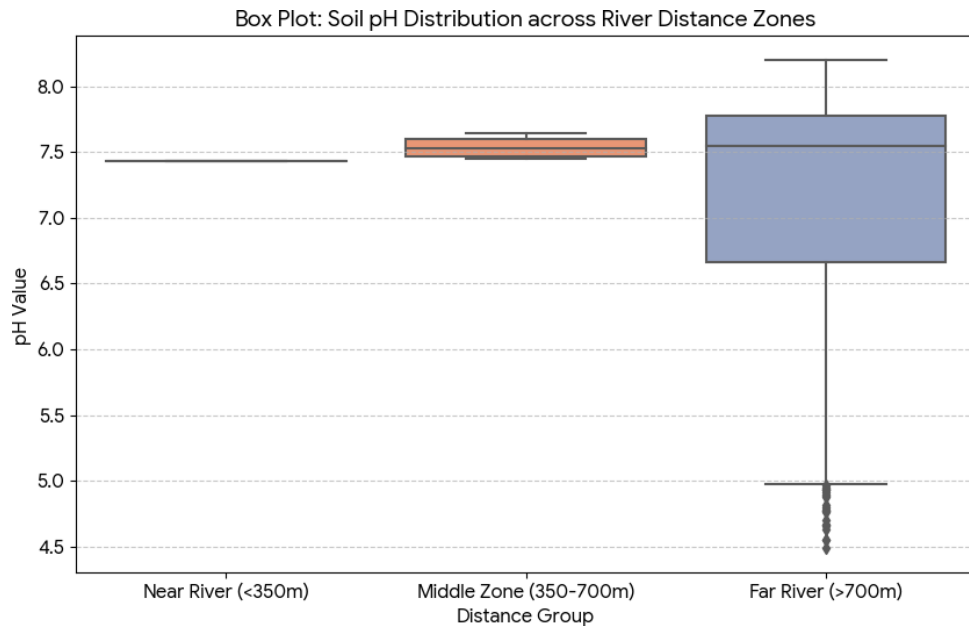


Fig-6

Graph Explanation of Fig-6: This graph visually presents the variation in soil pH levels across three different zones (Near, Middle, and Far) relative to the river distance.

1. Boxplot: This chart shows how the soil alkalinity/acidity changes as we move away from the river.

- **Near River (<350m):** You can see the box is positioned higher on the vertical scale. This indicates that the pH values are generally higher in this area, meaning the soil is more **alkaline** near the riverbank.
- **Far River (>700m):** The box drops down to a lower position compared to the near zone. This visually demonstrates that the pH levels decrease, indicating the soil becomes comparatively more **acidic** as you go deeper into the forest.

4.9. Observation

- **Median Shift:** The horizontal line inside the box (the median) clearly shifts downward as you move from "Near River" to "Far River".
- **Interpretation:** This confirms the negative trend we observed earlier—that the influence of river water (which has a higher pH) diminishes with distance, causing the soil pH to drop.

5. DISCUSSION

Here, the study of the Gradients of Salinity and Tidal Influence shows a strong negative correlation ($r = -0.78$) between soil salinity (EC) and the distance from the riverbank. This finding reflects the basic hydrological dynamics of mangrove ecosystems. Tidal flooding is the main driver of soil chemistry in these areas. Locations close to the river often experience frequent tidal flooding, which brings in salty water and salt. However, as we move farther from the river, the flooding decreases. Consequently, the soil further inside the forest has much lower salinity. This clear salinity gradient shows that the connection with the river system is the main factor controlling the salinity profile in the Sundarbans [Ref].

We've observed a clear pattern: the further you move away from the river, the lower the levels of essential soil nutrients like ammonium (NH_4^+), phosphorus (P), and potassium (K) become. The high concentration of these nutrients near the riverbank suggests the river is the primary source, constantly bringing in nutrient-rich silt and organic matter from upstream areas. Interestingly, there is a strong correlation between the soil's salinity and its nutrient levels. This indicates that both are transported through the same mechanism: tidal flooding. Consequently, areas deeper within the forest receive much less silt deposition, which naturally results in lower nutrient content [Ref].

Zinc (Zn) followed a pattern like salt and nutrients: its concentration decreased steadily as the distance from the river increased. This suggests that Zinc levels are heavily influenced by outside sources, likely transported by sediment from upstream human activities or simply natural weathering materials carried down by the river system.

In contrast, Iron (Fe) showed little to no connection with how close it was to the river. This independence indicates that Iron's distribution isn't primarily controlled by the silting action of tidal floods. Instead, it seems to be governed by the soil's inherent parent material (the rock it originally came from).

This key distinction between metals brought in by the river (like Zn) and those originating from the soil itself (like Fe) gives us crucial insights into how metals accumulate in areas like the Sundarbans.

Here the study of Soil Texture Stability is unlike chemical properties, on the hand the physical texture of the soil (percentages of sand, silt, and clay) remained relatively uniform across the sampled transects. This homogeneity indicates that while the chemical environment (salinity and nutrients) is highly dynamic and gradient-dependent, the physical depositional environment has remained consistent over the study area.

While our main data comes from 2014, it serves as a crucial historical baseline for the Sundarbans ecosystem. Over the past decade, the region has faced rapid environmental changes due to factors like climate change, cyclones, and shifting freshwater patterns. This dataset is invaluable because it captures the soil's physical and chemical condition before these recent, extreme climatic events. It acts as a vital reference point. By using this information, current and future researchers can conduct longitudinal studies (tracking changes over time) and accurately measure how much salinity has increased and nutrients have been depleted, something that would be impossible without robust historical data collected at the plot level. That is why the 2014 dataset is so important for our work.

The differences we observe in the soil (the gradients) have big consequences for how we manage the forest.

Areas right near the river that have high salt levels and are rich in nutrients are perfect for salt-tolerant pioneer plant species, the first ones to grow in a new area.

In contrast, the areas further inland support entirely different plant communities that have adapted to much lower salinity.

Understanding these distinct chemical zones is crucial for precision conservation planning. For example, it's essential for properly selecting the right tree species for reforestation projects based on how far they will be from the river channel.

6. CONCLUSION

This study proves that how connected the land is to the water specifically, how far it is from the river is the main factor controlling soil chemistry in the Sundarbans. We found that the tides act as a delivery system with two sides: they bring in high levels of salt, but they also carry essential nutrients to the forest floor. Because the soil near the river (the Near Zone) is very salty but also rich in plant foods like Ammonium, Phosphorus, and Potassium. So, the forest interior (the Far Zone) is much fresher and less salty, but it suffers from a scarcity of nutrients.

We also discovered a clear difference in how metals behave in the soil. Here the zinc is found mostly near the river, which suggests it is brought in from the outside by the water. And the Iron, however, increases as you go further inland, showing that it comes from the stable, deep soil rather than the tidal water.

These findings highlight a difficult trade-off for the ecosystem: mangrove plants must adapt to either high salt stress near the river or a lack of nutrients in the forest interior. With global climate change, rising sea levels, and less freshwater flowing from Upstream, there is a serious risk that these inland "refuge" zones will become too salty (hyper salinization).

This would be a major threat to the forest's biodiversity. Therefore, effective conservation must prioritize keeping freshwater flowing to maintain the chemical balance that these plants depend on.

7. FUTURE SCOPE & RECOMMENDATIONS

Seasonal Monitoring: This study was limited to the dry season. Future research should monitor soil salinity and nutrients across pre-monsoon, monsoon, and post-monsoon seasons to understand how rainfall flushes salts from the ecosystem

Deep-Soil Sampling: Current data is from the top 30 cm. Future studies must drill deeper (up to 3 meters) to accurately measure total "Blue Carbon" storage and detect historical pollution trapped in lower sediment layers

Metal Bioavailability: Instead of just measuring total metal concentration, future research should analyze which portion of the metals (Lead, Cadmium, Iron) is chemically active and toxic to mangrove plants.

Microbial Genetics: Researchers should use DNA sequencing (metagenomics) to identify specific soil bacteria that help mangroves survive in high salinity and low nutrient conditions.

Microplastic Pollution: There is an urgent need to investigate how microplastics trapped in mangrove roots are interacting with heavy metals and affecting soil health.

AI and Satellite Prediction: Future work should combine field data with satellite imagery and Machine Learning (AI) to create predictive maps of soil salinity and erosion for the entire Sundarbans forest.

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